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ORIGINAL RESEARCH ARTICLE

# Budget Impact of the LungFlag™ Predictive Risk Model for Lung Cancer Screening

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## Abstract

**Background/Objective** Lung cancer screening has historically been associated with false-positive results that can lead to complications and increased costs. LungFlag is a machine-learning risk prediction model using individual-level data to identify people at high risk of developing non-small cell lung cancer (NSCLC), prompting healthcare professionals to consider appropriate actions such as recommended screening with low-dose computed tomography. This study evaluated the budget impact of using the LungFlag risk prediction model to identify candidates for lung cancer screening from a US payer perspective.

**Methods** This budget impact model estimated total annual costs with or without LungFlag for a hypothetical 1 million-member US commercial health plan. Incremental costs were evaluated over a 5-year period with a cycle length of 1 year; no discounting was applied. All costs were adjusted to 2021 US dollars. The population included screening-naïve individuals aged 50–80 years with a ≥20-pack-years smoking history who were either current smokers or former smokers who had quit smoking in the previous 15 years (US Preventive Services Task Force 2021 criteria). LungFlag performance was measured 9–12 months before clinical diagnosis. Sensitivity and scenario analyses explored the robustness of base-case assumptions.

**Results** The budget impact model showed that LungFlag was associated with a total cumulative cost savings of US$2.8 million over 5 years, largely attributable to reduced advanced NSCLC treatment costs (−$4.4 million) as more patients were identified with early stage I rather than stage III/IV disease. Using LungFlag, an estimated 17 additional NSCLC diagnoses were identified, 22 fewer NSCLC-related deaths occurred over 5 years, 50 more patients had stage I NSCLC at diagnosis, and 33 fewer patients had stage III or IV NSCLC at diagnosis.

**Conclusions** Using LungFlag to identify candidates for lung cancer screening was estimated to save US$2.8 million over 5 years from a US commercial health plan perspective, largely attributable to reduced costs of treating advanced NSCLC.

## 1 Introduction

The prevalence of lung cancer and lung cancer–associated mortality continues to increase worldwide [1]. Earlier detection of lung cancer through screening programs and by addressing modifiable risk factors are effective evidence-based strategies to reduce the burden of lung cancer [2]. The US Preventive Services Task Force (USPSTF) 2021 guidelines recommend annual screening for lung cancer with low-dose computed tomography (LDCT) for individuals aged 50–80 years with no history of non-small cell lung cancer (NSCLC) who are current smokers or have quit smoking in the last 15 years and have a smoking history of at least 20 pack years [3]. LDCT for early detection has been shown to reduce mortality [4, 5] and is recommended by the USPSTF and several prominent clinical practice guidelines authorities [3, 6–8]. Indeed, long-term follow-up from the US National Lung Screening Trial (NLST; median 11–12 years) showed that LDCT reduced both lung cancer–specific and all-cause mortality [9], and the Dutch-Belgian Nederlands-Leuvens Longkanker Screenings Onderzoek (NELSON) trial showed significantly fewer lung cancer–related deaths among both men and women receiving LDCT compared with no screening after 10 years of follow-up [10].

Historically, screening for people at high risk of lung cancer has also been associated with false-positive results that can lead to complications, inconvenience, and increased costs [11]. Moreover, emerging liquid biopsy approaches, including circulating tumor DNA, extracellular vesicles, and non-coding RNAs, show promise for early lung cancer

Extended author information available on the last page of the article

**Key Points for Decision Makers**

LungFlag is a machine-learning risk prediction model developed to identify people at high risk of developing non-small cell lung cancer (NSCLC) using individual-level data, prompting healthcare professionals to consider appropriate actions such as recommended screening with low-dose computed tomography.

Using LungFlag for prescreening of a US commercial health plan population yielded 17 additional NSCLC diagnoses, 22 fewer NSCLC-related deaths over 5 years, and 33 fewer patients with stage III/IV NSCLC at diagnosis.

LungFlag was associated with a total cumulative cost savings of US$2.8 million over 5 years, largely attributable to reduced advanced NSCLC treatment costs (−$4.4 million) as more patients were identified with stage I versus stage III/IV disease.

detection and prognostic monitoring, potentially complementing imaging-based screening [12]. Advances in artificial intelligence applied to CT scans have demonstrated improved sensitivity in nodule detection and classification, while also reducing false-positive rates [13]. Importantly, pragmatic strategies in primary care, such as electronic health record prompts, shared-decision tools, and integration with smoking cessation, have been identified as effective levers for increasing lung cancer screening uptake [14, 15].

LungFlag™ is a machine-learning–developed risk prediction model designed to identify people at high risk of developing NSCLC using individual-level data, prompting their healthcare professional to consider appropriate actions such as recommending screening with LDCT. Key predictors include smoking duration, age, pack-years, body mass index (BMI), white blood cell (WBC) count, history of chronic obstructive pulmonary disease (COPD), time since quitting smoking, Hispanic ethnicity, high-density lipoprotein trends over time, and red cell distribution width values. Longer smoking duration, older age, higher number of pack-years, higher WBC count, and history of COPD all contribute to a greater risk of cancer, while higher BMI, longer time since quitting, and Hispanic ethnicity contribute to a lower risk of cancer [16]. The algorithm was trained and validated using 2008–2015 data from the Kaiser Permanente Southern California Cancer Registry and Research Data Warehouse. By identifying patients with local-stage NSCLC, it may be possible to provide more timely, effective lung cancer treatment that could offer improved clinical outcomes and potentially reduce the overall costs of treatment. LungFlag has shown higher sensitivity and diagnostic odds ratios for predicting NSCLC diagnosis 9–12 months before clinical diagnosis compared with USPSTF screening guidelines and the 2012 Prostate, Lung, Colorectal and Ovarian (PLCOm2012) risk model when applied to individuals meeting USPSTF screening criteria [16].

This study evaluated the budget impact of using LungFlag to identify screening-naïve individuals at high risk of NSCLC who may be candidates for LDCT scans as an adjunct to existing screening guidelines from a US commercial payer perspective.

## 2 Methods

### 2.1 Model Overview

This budget impact model (BIM) was developed in Microsoft Excel® (www.microsoft.com) to estimate the costs associated with using LungFlag as an adjunct to the 2021 USPSTF screening guidelines for a hypothetical 1 million-member US commercial health plan. In this model, LungFlag was applied as a prescreening 'safety-net' intervention to identify screening-naïve individuals who met USPSTF eligibility criteria but had not undergone LDCT, reflecting the primary real-world gap the tool is intended to address. Patients flagged by LungFlag were offered LDCT screening per USPSTF guidelines. A schematic of how LungFlag integrates into the existing clinical workflow is provided in Fig. 1. The BIM calculates the total expected annual costs of NSCLC screening with LDCT with or without LungFlag, including costs of prescreening, LungFlag, LDCT screening (initial, follow-up, and total costs), and NSCLC diagnosis and treatment. Incremental costs were evaluated over a 5-year period with a cycle length of 1 year; no discounting was applied. All costs were adjusted to 2021 US dollars using the US Bureau of Labor Statistics Consumer Price Index for Medical Care [17]. A scalar (1.6) was applied to align healthcare costs with those expected for a commercial health plan based on the ratio of hospital payments for private insurers versus Medicare from the American Hospital Association 2020 TrendWatch ChartBook [18]. Results are expressed as total annual costs, total cumulative costs, and on a per-member-per-month basis.

### 2.2 Population

The base-case analysis assumed that LungFlag would be used as a 'safety net' to screen individuals who met the 2021 USPSTF criteria for lung cancer screening but had not been screened to identify individuals at highest risk. This included individuals 50–80 years old with a ≥20-pack-year smoking history who were current smokers or former smokers who had quit smoking in the previous 15 years [3]. The

age distribution of US private insurance plan members was based on the most recent health insurance data by type of coverage from the US Census Bureau (electronic supplementary material [ESM], Table S1) [19]. Overall lung cancer prevalence was estimated to be 0.34% and current or former smoking prevalence was estimated to be 46.3% among those aged 50–80 years based on estimates from the National Cancer Institute (NCI) and the US Centers for Disease Control and Prevention (CDC) [20, 21]. The NSCLC incidence rate for the USPSTF-eligible population was calculated to be 398 per 100,000 per year [22] based on age-specific estimates and the age distribution of the hypothetical health plan, where 59% of individuals were <45 years of age, smoking prevalence ranged from 44.7 to 45.2%, and lung cancer incidence increased with age from 14.4/100,000 (45–49 years) to 304.1/100,000 (≥85 years) (Table S1, ESM) [20, 21, 23, 24].

Overall stage distribution of incident NSCLC cases was based on data from the US NCI Surveillance, Epidemiology, and End Results (SEER) program [25]. NSCLC stage distribution among screening-detected NSCLC cases at the time of diagnosis was based on the stage distribution of incident cases from the NLST, providing screening-detected NSCLC stage estimates of 61.7% for stage I, 7.1% for stage II, and 15.6% for both stages III and IV (Table S2, ESM) [26].

## 2.3 Lung Cancer Screening Eligibility, Adherence, and Accuracy

One-quarter (25.5%) of ever smokers 50–80 years old were assumed to meet the 2021 USPSTF criteria based on published models for the USPSTF [27, 28]. Adherence to USPSTF screening recommendations for LDCT was assumed to be 12.5% for the reference scenario (no LungFlag) based on published estimates by the US CDC [29]. The sensitivity rate of 84.9% was based on a retrospective analysis of the LDCT cohort from the NLST [30], and the rate of positive results among those who underwent LDCT was 13.6% in the NLST [31]. Uptake of LDCT screening recommendations in the LungFlag cohort was assumed to be 68.1% based on the reported adherence rates to recommendations for colorectal cancer screening from a comparable algorithm [32]. The performance of LungFlag was measured in a timeframe of 9–12 months before clinical diagnosis, reflecting the predefined retrospective sampling window used in Gould et al. (2021) [16], where blood test dates were linked to each patient's subsequently recorded diagnosis date. At a cutoff representing a false-positive rate of 3%, the sensitivity was 19.2% (adapted from Gould et al. 2021) [16].

## 2.4 Costs of Low-Dose Computed Tomography (LDCT) and Non-small Cell Lung Cancer (NSCLC) Diagnosis

The modeled price for LungFlag included a one-time installation cost and an annual license fee. Costs related to LDCT screening and NSCLC diagnosis are provided in Table S3 (see ESM) [33–37]. All individuals undergoing screening with LDCT were assumed to have a prescreening visit, and those with positive initial findings had a diagnostic LDCT and an office visit. Those with positive findings on annual LDCT scan and those with clinically detected NSCLC were assumed to receive additional diagnostic testing and imaging. Unit costs were sourced from the Centers for Medicare & Medicaid Services Physician Fee Schedule [38].

## 2.5 NSCLC Probability of Survival and Treatment Costs

Annual probabilities of survival by NSCLC stage after the first year since NSCLC diagnosis were based on digitized

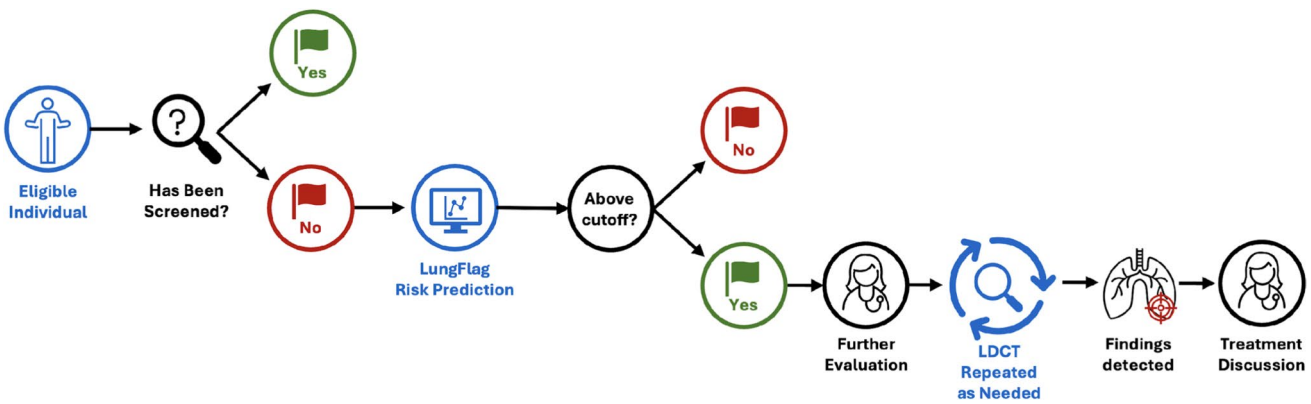



**Fig. 1** LungFlag integration in the existing clinical workflow. *LDCT* low-dose computed tomography

Kaplan-Meier survival curves from Önal et al. and from the SEER program, with 5-year estimates ranging from 97.4–65.0% for stage 1, 78.9–35.9% for stage II, 62.9–15.4% for stage III, and 32.5–2.5% for stage IV (Table S4, ESM) [21, 39].

Stage-specific annual costs of NSCLC treatment were based on Sheehan et al. (Table S5, ESM) [40]. NSCLC treatment costs were considered for an initial phase (first 6 months from diagnosis), a continuing phase (with any costs associated with recurrences or disease progression), and end of life (last 6 months of life). Continuing phase costs for all stages were applied annually to the portion of model cohorts surviving after the first year of model entry. End-of-life costs were applied to the total number of estimated deaths each year. Treatment costs for patients with NSCLC were augmented with estimates of the increase in outpatient systemic therapy costs from 2013 to 2018 [41]. A summary of key inputs is provided in Table 1.

### 2.6 Sensitivity and Scenario Analyses

One-way sensitivity analyses considered ±25% variations in the base case values for adherence to USPSTF screening, adherence to LungFlag screening, screening costs, diagnostic procedure costs, and NSCLC treatment costs. Scenario analyses were performed to evaluate the results for a screening-eligible population that was different from the USPSTF guideline (adults aged 50–80 years with a ≥20-pack-year smoking history, current or former smokers who had quit smoking in the previous 15 years), including ever-smokers either 50–80 years old (as with USPSTF criteria) or 45–90 years old. Scenario analyses also considered LungFlag false-positive rates of 5 and 10% (as opposed to 3% in the base-case analysis), with corresponding sensitivity increased to 26 and 38%, respectively, for both the base-case screening-eligible and the ever-smoker populations.

## 3 Results

### 3.1 Base-Case Results: LDCT Screening, NSCLC Diagnoses, and Deaths

In a hypothetical US commercial health plan, using LungFlag was estimated to provide LDCT screening for an additional 675 members per year (5275 per year with LungFlag; 4600 per year without LungFlag) (Table 2). Each year, using LungFlag resulted in more screening-detected NSCLC cases (32 vs 16 per year) and 67 fewer clinically detected cases compared with the No LungFlag cohort (year 1, 131 each; years 2–5, 114 vs 131, respectively). Using LungFlag resulted in more NSCLC cases being diagnosed at stage I and fewer diagnoses at stage III or IV than in the No LungFlag cohort (Fig. 2).

Overall, 36,803 (4%) out of 1 million members per year were estimated to be aged 50–80 years, without a history of lung cancer, and eligible for screening based on USPSTF criteria. Of these 36,803 members, 4600 (12%) were expected to be screened for lung cancer, leaving 32,203 (88%) screening-naïve members as candidates for prescreening with LungFlag. Over 5 years, using LungFlag was estimated to identify 3371 additional members for LDCT screening, with 17 additional NSCLC diagnoses (84 screening detected; 67 clinically detected) and 22 fewer NSCLC-related deaths (Table 2). When analyzed by stage, using LungFlag was estimated to yield 50 more NSCLC diagnoses at stage I and 33 fewer at stage III or IV (Fig. 2).

**Table 1** Summary of key model inputs

| Model input | Value | Source(s) |
|---|---|---|
| Perspective | US commercial health plan | Assumption |
| Currency, costing year | 2021 US dollars with 1.6 scalar | US BLS [17]<br>AHA 2020 Trend-Watch Chart-Book [18] |
| Population size | 1 million health plan members | Assumption |
| Screening-eligible population (base case) | Adults 50–80 years old with a ≥20-pack-year smoking history who were either current or former smokers who had quit smoking in the previous 15 years | USPSTF [3] |
| Adherence to LDCT screening once flagged by LungFlag | 68.1% | Goshen et al. [32] |
| LungFlag false-positive rate | 3% | Gould et al. [16] |
| NSCLC incidence rate per year in the target population | 398 per 100,000 | Tindle et al. [22] |

*AHA* American Hospital Association, *LDCT* low-dose computed tomography, *NSCLC* non-small cell lung cancer, *US BLS* United States Bureau of Labor Statistics, *USPSTF* United States Preventive Services Task Force

### 3.2 Base Case Results: Budget Impact of Using LungFlag

LungFlag was associated with increased annual costs in Years 1 and 2 but reduced costs in Years 3–5 (Table 3). Overall, LungFlag was associated with total cumulative cost savings of US$2.83 million over 5 years. Overall cost savings from using LungFlag were largely attributable to reductions in the costs of advanced NSCLC treatment (–US$4.38 million) from more patients being identified with earlier-stage disease.

### 3.3 Sensitivity and Scenario Analyses

One-way sensitivity analyses showed cost savings across all of the scenarios exploring ±25% variations in base-case parameters for screening adherence and cost inputs (Table S6, ESM). Overall, cumulative 5-year cost savings with LungFlag were generally observed across the scenario

**Table 2** Estimated LDCT screening, NSCLC diagnoses and deaths with and without LungFlag

| Number of individuals, n | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Total[a] | Difference, LungFlag vs no LungFlag |
|---|---|---|---|---|---|---|---|
| Individuals screened with LDCT plus follow-up LDCT[b] | | | | | | | |
| LungFlag | 5275 | 5275 | 5275 | 5275 | 5275 | 26,373 | 3371 |
| No LungFlag | 4600 | 4600 | 4600 | 4600 | 4600 | 23,000 | |
| Screening detected NSCLC | | | | | | | |
| LungFlag | 32 | 32 | 32 | 32 | 32 | 162 | 84 |
| No LungFlag | 16 | 16 | 16 | 16 | 16 | 78 | |
| Clinically detected NSCLC | | | | | | | |
| LungFlag | 131 | 114 | 114 | 114 | 114 | 588 | −67 |
| No LungFlag | 131 | 131 | 131 | 131 | 131 | 655 | |
| NSCLC deaths | | | | | | | |
| LungFlag | 0 | 74 | 112 | 122 | 127 | 434 | −22 |
| No LungFlag | 0 | 74 | 118 | 131 | 135 | 457 | |

*LDCT* low-dose computed tomography, *NSCLC* non-small cell lung cancer

[a]Total results do not calculate exactly from annual estimates due to rounding

[b]Estimates include 4600 members compliant with screening, with 627 with positive LDCT requiring confirmation in the No LungFlag group ($n = 5227$), and 674 additional members for screening plus 91 with positive LDCT requiring follow-up in the LungFlag group ($n = 5992$)

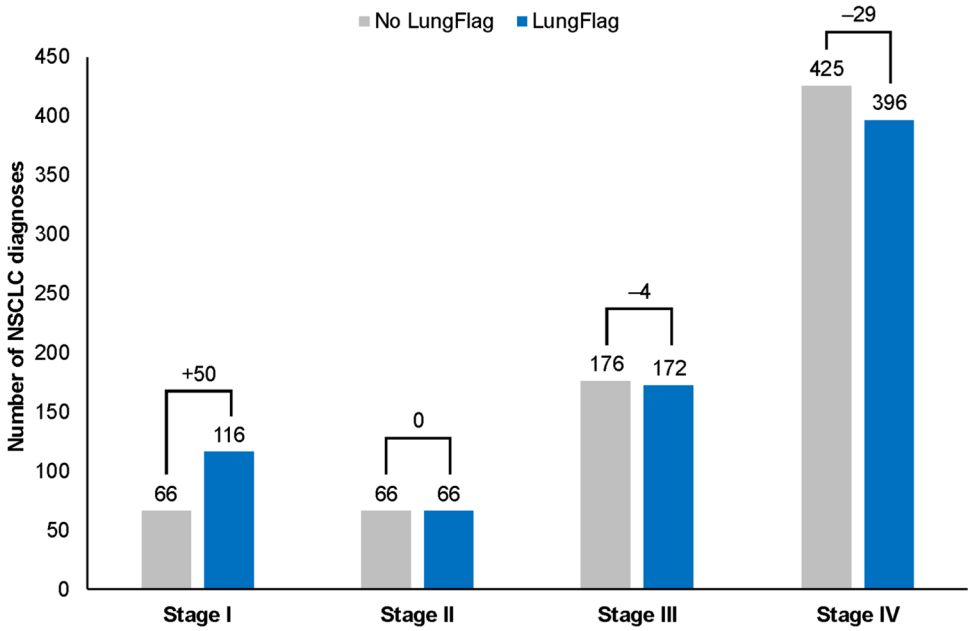



**Fig. 2** NSCLC diagnoses over 5 years with and without LungFlag, by stage. *NSCLC* non-small cell lung cancer

analyses. Cost savings were observed with ever-smokers aged 50–80 years or 45–90 years and with the alternative LungFlag false-positive rates of 5 and 10%, with the exception of 10% false-positive rates applied to ever-smokers aged 45–90 years (cumulative budget impact increased to US$6.46 million) or ever-smokers aged 50–80 years (US$3.64 million) (Table S6, ESM).

## 4 Discussion

This budget impact model estimated substantial cost savings when LungFlag was used to identify lung cancer screening–eligible members of a US commercial health plan. Annually, approximately 88% of health plan members aged 50–80 years who met USPSTF-recommended screening criteria were candidates for prescreening with LungFlag. Over 5 years, an additional 3371 plan members were screened for NSCLC, resulting in 84 additional NSCLC diagnoses made via screening and the prevention of an estimated 22 NSCLC-related deaths. Notably, using LungFlag identified 52 additional patients with early stage I rather than stage III or IV NSCLC, which contributed to downstream cost savings by avoiding late-stage NSCLC treatment. Overall, use of LungFlag was estimated to save US$2.8 million over 5 years, where cost savings are realized starting in Year 3. These cost savings are largely attributable to reduced costs related to treating earlier-stage rather than advanced-stage NSCLC, where treatment costs are higher [42]. Scenario analyses exploring variations in the base case assumptions also showed cost savings when LungFlag was used in expanded prescreening populations by age and ever-smoker status, with a few specific exceptions. Taken together, this BIM suggests that using LungFlag to prescreen individuals who have not received lung cancer screening aligned with USPSTF recommendations would have a positive health economic impact to a US commercial health plan.

Findings from this study should be considered in the context of certain strengths and limitations. LungFlag is a novel machine learning–developed risk prediction tool that uses individual-level data, offering a precise approach to addressing historical challenges to lung cancer screening and diagnosis, which can have a substantial impact on downstream clinical and economic outcomes. LungFlag has demonstrated improved rates of NSCLC detection compared with USPSTF recommendations and the PLCOm2012 risk model [16]. However, the LungFlag false-positive rate and adherence to screening opportunities should be considered in real-world applications. Since the accuracy of LungFlag was based on a retrospective case-control study, it should be noted that estimates of the sensitivity and specificity of LungFlag are only known in patients with clinically diagnosed disease. The assumption for adherence to LungFlag screening recommendations was based on a study of a similar screening approach for colon cancer in Israel, which may not reflect real-world adherence to LungFlag screening recommendations in the US. However, LungFlag would be expected to provide cost savings even if the adherence to LungFlag screening recommendations was assumed to be similar to adherence to USPSTF recommendations. Some assumptions related to NSCLC treatment costs in this model had to be made based on the available data and the ever-changing NSCLC treatment landscape. However, some of the more recently approved adjuvant NSCLC treatments have been included in the early treatment costs. It should be noted that while we based the age distribution for the commercially insured population on a published estimate of health insurance coverage in the US, some costs covered by public insurance programs (such as Medicare for eligible patients) would not have been captured. In this way, the costs of care may have been underestimated which, in turn, may have underestimated the cost savings estimated with the adoption of LungFlag in this population. All cost estimates are presented in 2021 US dollars, and results should be

**Table 3** Differences in costs with versus without LungFlag: base-case budget impact results

| Incremental costs, US$ | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Cumulative |
|---|---|---|---|---|---|---|
| Prescreening and screening costs | 315,380 | 285,380 | 285,380 | 285,380 | 285,380 | 1,456,901 |
| Diagnostic procedures | 28,299 | 23,130 | 23,130 | 23,130 | 23,130 | 120,820 |
| NSCLC treatment | 2,540,257 | 186,468 | −1,864,645 | −2,635,390 | −2,637,025 | −4,410,335 |
| Total budget impact | 2,883,936 | 494,979 | −1,556,134 | −2,326,880 | −2,328,515 | −2,832,614 |
| Cumulative budget impact, 5 years | 2,883,936 | 3,378,915 | 1,822,781 | –504,099 | −2,832,614 | |
| Incremental budget impact per member per month | 0.24 | 0.04 | −0.13 | −0.19 | −0.19 | |

Note: Negative cost values indicate cost savings. Cumulative budget impact over 5 years represents the accumulation of costs and savings according to the annual total budget impact calculations

*NSCLC* non-small cell lung cancer

interpreted with consideration that more recent cost changes may not be reflected.

## 5 Conclusions

The adoption of LungFlag as an adjunct to existing screening guidelines for USPSTF-eligible individuals was estimated to prevent 22 NSCLC-related deaths with cost savings of approximately US$2.8 million over 5 years from a US commercial payer perspective in a hypothetical health plan with 1 million members. Higher initial costs with LungFlag were offset over subsequent years by reduced costs of treating advanced NSCLC driven by fewer patients being diagnosed with late-stage NSCLC when LungFlag was used. While these findings support LungFlag implementation to reduce cost burden, further real-world study is needed as results are likely to vary based on health system–specific structural and population factors. Overall, this study suggests that use of LungFlag in a US commercial health plan population would contribute to improved screening and diagnosis of NSCLC associated with reduced costs to the health plan over 5 years.

**Supplementary Information** The online version contains supplementary material available at https://doi.org/10.1007/s41669-025-00627-6.

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## Declarations

**Conflict of interest** MKG reports research funding from F. Hoffmann-La Roche Ltd; EC reports employment from F. Hoffmann-La Roche, which received research funding for this project from the study sponsor; NO reports employment/stockholdings for F. Hoffmann-La Roche Ltd; MO reports employment/stockholdings for F. Hoffmann-La Roche Ltd; SO reports employment/stockholdings for Genentech, the study sponsor; CHS reports employment/stockholdings for F. Hoffmann-La Roche Ltd.

**Ethics approval** Not applicable.

**Consent to participate** Not applicable.

**Consent for publication** Not applicable.

**Code availability** Formulae used in this model may be available to qualified researchers upon reasonable request.

**Data availability statement** All data supporting the findings of this study are available within the paper and its supplementary information.

**Author contributions** All authors contributed to the study design, interpretation of findings, critical review and revision of the manuscript, and approval of the manuscript for submission; CH, EC, NO, SO, and MO also contributed to acquisition of data, data analysis, and project management.

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## References

1. Sung H, Ferlay J, Siegel RL, Laversanne M, Soerjomataram I, Jemal A, et al. Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J Clin. 2021;71(3):209–49.
2. Oliver AL. Lung cancer: epidemiology and screening. Surg Clin North Am. 2022;102(3):335–44.
3. US Preventive Services Task Force, Krist AH, Davidson KW, Mangione CM, Barry MJ, Cabana M, et al. Screening for lung cancer: US Preventive Services Task Force Recommendation Statement. JAMA. 2021;325(10):962–70.
4. Bonney A, Malouf R, Marchal C, Manners D, Fong KM, Marshall HM, et al. Impact of low-dose computed tomography (LDCT) screening on lung cancer-related mortality. Cochrane Database Syst Rev. 2022;8(8):CD013829.
5. Lancaster HL, Heuvelmans MA, Oudkerk M. Low-dose computed tomography lung cancer screening: clinical evidence and implementation research. J Intern Med. 2022;292(1):68–80.
6. Jaklitsch MT, Jacobson FL, Austin JH, Field JK, Jett JR, Keshavjee S, et al. The American Association for Thoracic Surgery guidelines for lung cancer screening using low-dose computed tomography scans for lung cancer survivors and other high-risk groups. J Thorac Cardiovasc Surg. 2012;144(1):33–8.
7. Mazzone PJ, Silvestri GA, Souter LH, Caverly TJ, Kanne JP, Katki HA, et al. Screening for lung cancer: CHEST guideline and expert panel report. Chest. 2021;160(5):e427–94.
8. Wender R, Fontham ET, Barrera E Jr, Colditz GA, Church TR, Ettinger DS, et al. American Cancer Society lung cancer screening guidelines. CA Cancer J Clin. 2013;63(2):107–17.
9. National Lung Screening Trial Research Team. Lung cancer incidence and mortality with extended follow-up in the National Lung Screening Trial. J Thorac Oncol. 2019;14(10):1732–42.
10. de Koning HJ, van der Aalst CM, de Jong PA, Scholten ET, Nackaerts K, Heuvelmans MA, et al. Reduced lung-cancer mortality with volume CT screening in a randomized trial. N Engl J Med. 2020;382(6):503–13.
11. Jonas DE, Reuland DS, Reddy SM, Nagle M, Clark SD, Weber RP, et al. Screening for lung cancer with low-dose computed tomography: updated evidence report and systematic review for the US Preventive Services Task Force. JAMA. 2021;325(10):971–87.
12. Ren F, Fei Q, Qiu K, Zhang Y, Zhang H, Sun L. Liquid biopsy techniques and lung cancer: diagnosis, monitoring and evaluation. J Exp Clin Cancer Res. 2024;43(1):96.

13. Quanyang W, Yao H, Sicong W, Linlin Q, Zewei Z, Donghui H, et al. Artificial intelligence in lung cancer screening: detection, classification, prediction, and prognosis. Cancer Med. 2024;13(7):e7140.
14. Nunez ER, Ito Fukunaga M, Stevens GA, Yang JK, Reid SE, Spiegel JL, et al. Review of interventions that improve uptake of lung cancer screening: a cataloging of strategies that have been shown to work (or not). Chest. 2024;166(3):632–48.
15. Symvoulakis EK, Bouloukaki I, Christodoulakis A, Aravantinou-Karlatou A, Tsiligianni I. Optimal pathways to lung cancer screening in primary care settings: a scoping review. Curr Oncol. 2024;32(1):8.
16. Gould MK, Huang BZ, Tammemagi MC, Kinar Y, Shiff R. Machine learning for early lung cancer identification using routine clinical and laboratory data. Am J Respir Crit Care Med. 2021;204(4):445–53.
17. US Bureau of Labor Statistics. Consumer price index. 2024. [May 1, 2024]. Available from: https://www.bls.gov/cpi/data.htm.
18. American Hospital Association. Table 4.4: Aggregate hospital payment-to-cost ratios for private payers, Medicare, and Medicaid, 1995–2018. 2024. [September 24, 2024]. Available from: https://www.aha.org/system/files/2018-05/2018-chartbook-table-4-4.pdf.
19. Keisler-Starkey K, Bunch LN, Lindstrom RA. Health insurance coverage in the United States: 2022. [December 12, 2025]. Available from: https://www.census.gov/library/publications/2023/demo/p60-281.html.
20. Centers for Medicare & Medicaid Services. Physician fee schedule. 2024. [December 6, 2024]. Available from: https://www.cms.gov/medicare/payment/fee-schedules/physician.
21. National Cancer Institute. Surveillance, epidemiology, and end results program. 2024. [May 1, 2024]. Available from: https://seer.cancer.gov/statistics-network/explorer/.
22. Tindle HA, Stevenson Duncan M, Greevy RA, Vasan RS, Kundu S, Massion PP, et al. Lifetime smoking history and risk of lung cancer: results from the Framingham heart study. J Natl Cancer Inst. 2018;110(11):1201–7.
23. Keisler-Starkey K, Bunch LN. U.S. Census Bureau Current Population Reports, P60-274, Health Insurance Coverage in the United States: 2020. Washington, DC, 2021.
24. National Cancer Institute. SEER cancer statistics review (CSR) 1975-2015. 2024. [May 1, 2024]. Available from: https://seer.cancer.gov/csr/1975_2015/.
25. National Cancer Institute. Cancer stat facts: lung and bronchus cancer; 2021. 2021. [May 1, 2024]. Available from: https://seer.cancer.gov/statfacts/html/lungb.html.
26. Pinsky PF, Church TR, Izmirlian G, Kramer BS. The National Lung Screening Trial: results stratified by demographics, smoking history, and lung cancer histology. Cancer. 2013;119(22):3976–83.
27. Katki HA, Kovalchik SA, Berg CD, Cheung LC, Chaturvedi AK. Development and validation of risk models to select ever-smokers for CT lung cancer screening. JAMA. 2016;315(21):2300–11.
28. Meza R, Jeon J, Toumazis I, Ten Haaf K, Cao P, Bastani M, et al. Evaluation of the benefits and harms of lung cancer screening with low-dose computed tomography: modeling study for the US Preventive Services Task Force. JAMA. 2021;325(10):988–97.
29. Richards TB, Soman A, Thomas CC, VanFrank B, Henley SJ, Gallaway MS, et al. Screening for lung cancer—10 states, 2017. MMWR Morb Mortal Wkly Rep. 2020;69(8):201–6.
30. Pinsky PF, Gierada DS, Black W, Munden R, Nath H, Aberle D, et al. Performance of Lung-RADS in the National Lung Screening Trial: a retrospective assessment. Ann Intern Med. 2015;162(7):485–91.
31. Henderson LM, Bacchus L, Benefield T, Huamani Velasquez R, Rivera MP. Rates of positive lung cancer screening examinations in academic versus community practice. Transl Lung Cancer Res. 2020;9(4):1528–32.
32. Goshen R, Choman E, Ran A, Muller E, Kariv R, Chodick G, et al. Computer-assisted flagging of individuals at high risk of colorectal cancer in a large health maintenance organization using the ColonFlag test. JCO Clin Cancer Inform. 2018;2:1–8.
33. National Lung Screening Trial Research T, Aberle DR, Adams AM, Berg CD, Black WC, Clapp JD, et al. Reduced lung-cancer mortality with low-dose computed tomographic screening. N Engl J Med. 2011;365(5):395–409.
34. Roth JA, Sullivan SD, Goulart BH, Ravelo A, Sanderson JC, Ramsey SD. Projected clinical, resource use, and fiscal impacts of implementing low-dose computed tomography lung cancer screening in Medicare. J Oncol Pract. 2015;11(4):267–72.
35. Toumazis I, de Nijs K, Cao P, Bastani M, Munshi V, Ten Haaf K, et al. Cost-effectiveness evaluation of the 2021 US Preventive Services Task Force recommendation for lung cancer screening. JAMA Oncol. 2021;7(12):1833–42.
36. Toumazis I, Erdogan SA, Bastani M, Leung A, Plevritis SK. A cost-effectiveness analysis of lung cancer screening with low-dose computed tomography and a diagnostic biomarker. JNCI Cancer Spectr. 2021. https://doi.org/10.1093/jncics/pkab081.
37. Villanti AC, Jiang Y, Abrams DB, Pyenson BS. A cost-utility analysis of lung cancer screening and the additional benefits of incorporating smoking cessation interventions. PLoS One. 2013;8(8):e71379.
38. Centers for Medicare & Medicaid Services. Physician fee schedule 2022. 2022. [September 23, 2022]. Available from: https://www.cms.gov/Medicare/Medicare-Fee-for-Service-Payment/PhysicianFeeSched/.
39. Onal O, Kocer M, Eroglu HN, Yilmaz SD, Eroglu I, Karadogan D. Survival analysis and factors affecting survival in patients who presented to the medical oncology unit with non-small cell lung cancer. Turk J Med Sci. 2020;50(8):1838–50.
40. Sheehan DF, Criss SD, Chen Y, Eckel A, Palazzo L, Tramontano AC, et al. Lung cancer costs by treatment strategy and phase of care among patients enrolled in Medicare. Cancer Med. 2019;8(1):94–103.
41. Zhang X, Beachler DC, Masters E, Liu F, Yang M, Dinh J, et al. Health care resource utilization and costs associated with advanced or metastatic nonsmall cell lung cancer in the United States. J Manag Care Spec Pharm. 2022;28(2):255–65.
42. Jovanoski N, Abogunrin S, Di Maio D, Belleli R, Hudson P, Bhadti S, et al. Systematic literature review to identify cost and resource use data in patients with early-stage non-small cell lung cancer (NSCLC). Pharmacoeconomics. 2023;41(11):1437–52.

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